

CENTRAL KENTUCKY, KY, USA

WMO: 720455

Lat: 37.633N

Lon: 84.333W

Elev: 305

StdP: 97.71

Time Zone: -5.00 (NAE)

Period: 11-19

WBAN: 00145

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.0	-10.1	-18.9	0.7	-10.7	-16.1	0.9	-9.2	11.0	5.6	10.0	3.7	1.7	330	0.416

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	32.9	23.1	32.1	22.8	30.9	22.6	24.8	30.2	24.1	29.2	23.5	28.5	2.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	18.3	27.6	22.5	17.8	27.2	22.2	17.5	26.9	76.7	30.2	73.6	29.5	71.6	28.4	27.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.9	7.8	6.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			-17.1	26.0	5.0	0.7	-20.7	26.5	-23.7	26.9	-26.5	27.3	-30.1	27.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.8	1.3	4.1	7.8	14.2	19.2	22.9	24.6	23.3	20.5	14.4	8.0	5.1		
	DBStd	9.45	7.08	6.91	6.39	5.24	4.63	2.92	2.71	2.42	4.10	5.09	5.49	5.71		
	HDD10.0	898	279	186	119	22	1	0	0	0	0	17	101	174		
	HDD18.3	2338	527	399	327	142	49	3	1	1	24	142	313	411		
	CDD10.0	2298	11	21	52	147	286	386	452	413	316	154	39	22		
	CDD18.3	697	0	1	2	17	76	140	194	156	90	20	2	0		
	CDH23.3	5849	0	1	13	134	568	1192	1717	1261	799	158	7	0		
	CDH26.7	1979	0	0	0	13	139	403	679	417	298	31	1	0		
(14) Wind	WSAvg	2.6	3.3	3.2	3.2	3.3	2.6	2.2	1.7	1.6	1.7	2.5	2.7	3.0		
(15) Precipitation	PrecAvg	1193	76	77	110	102	125	116	117	104	95	76	93	99		
	PrecMax	1614	203	215	268	275	252	243	227	240	254	183	227	328		
	PrecMin	831	14	9	25	24	24	20	27	29	15	0	19	14		
	PrecStd	209	42	46	60	53	54	54	46	51	55	42	45	54		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.7	22.4	25.9	27.9	31.3	34.0	37.0	33.7	35.0	29.8	23.9	20.1		
		MCWB	14.9	16.2	17.0	18.2	22.1	21.5	22.8	22.4	22.2	21.8	16.9	16.0		
	2%	DB	16.4	19.8	22.7	27.0	29.1	32.1	33.0	32.1	32.4	27.3	20.9	18.0		
		MCWB	12.7	15.2	15.0	17.6	21.0	22.3	24.1	22.0	22.0	19.6	15.4	14.3		
	5%	DB	14.7	17.2	20.2	25.0	27.9	30.8	32.1	30.8	30.1	25.1	18.0	16.2		
		MCWB	11.7	12.8	14.0	16.6	20.2	22.1	24.1	22.6	21.7	18.1	13.3	13.1		
	10%	DB	12.3	14.8	17.8	22.8	27.1	29.0	30.2	29.0	27.8	22.7	16.9	14.1		
		MCWB	9.4	11.0	12.2	15.6	19.7	21.7	23.2	21.9	21.2	17.3	12.0	11.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.6	17.1	18.0	20.2	23.3	24.7	26.2	24.7	24.2	22.7	17.8	17.0		
		MCDB	17.9	20.8	24.5	25.9	29.2	30.2	31.5	29.7	29.7	27.9	21.7	19.4		
	2%	WB	14.0	15.6	16.4	19.0	22.1	23.8	25.3	24.1	23.3	21.3	16.7	15.2		
		MCDB	16.1	18.6	21.1	25.1	27.9	29.4	31.1	29.2	29.3	24.7	19.7	17.5		
	5%	WB	12.0	14.1	15.1	17.9	21.4	23.0	24.6	23.4	22.5	19.8	14.8	13.6		
		MCDB	14.3	16.6	19.2	23.0	26.8	28.6	29.9	28.3	28.5	23.3	17.5	15.9		
	10%	WB	9.8	11.9	13.7	16.8	20.4	22.3	23.9	22.8	21.8	18.0	12.7	12.1		
		MCDB	12.2	13.8	17.2	21.5	25.3	27.6	28.8	27.4	27.0	22.5	15.7	14.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.1	9.5	10.3	11.4	10.3	10.8	10.2	10.6	11.2	10.9	10.4	8.3		
		MCDBR	10.8	11.7	12.3	13.5	11.4	12.8	11.6	12.5	13.9	13.1	10.9	10.6		
	5% WB	MCWBR	8.4	8.3	7.6	7.0	5.3	5.2	4.2	5.0	5.1	6.2	6.9	8.2		
		MCWBR	10.4	11.1	10.4	11.0	10.3	10.4	9.9	9.7	11.4	10.9	8.6	9.5		
(40) Clear-Sky Solar Irradiance	taub	taub	0.307	0.318	0.348	0.391	0.441	0.450	0.477	0.476	0.406	0.355	0.319	0.299		
		taud	2.478	2.431	2.400	2.280	2.177	2.189	2.130	2.123	2.293	2.431	2.492	2.510		
	Ebn at Noon	Ebn at Noon	881	914	914	889	846	835	809	800	841	859	858	866		
		Edh at Noon	81	96	109	131	148	146	154	151	121	95	80	74		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.95	2.60	3.73	4.96	5.55	6.17	5.79	5.46	4.53	3.43	2.40	1.71		
	RadStd	RadStd	0.22	0.38	0.32	0.37	0.52	0.33	0.30	0.30	0.48	0.42	0.26	0.17		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (23 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+120	+74

Nomenclature: See separate page